Arian Salari

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Education

University of California, Irvine — **Irvine, CA**
B.S. in Mechanical Engineering, Honors Collegium — *September 2026 – June 2028*

- Coursework: Statics, Dynamics, Thermodynamics, Electric Circuits, Materials Science

Irvine Valley College — **Irvine, CA**
Associate's in Engineering, Honors Program, Cumulative GPA: 3.95 — *August 2024 – May 2026*

- Coursework: CAD Techniques, Physics 4A/4B/4C, Calculus I/II/III, Linear Algebra, Differential Equations, Computer Programming

Experience

Rezvani Motors — **Jan 2025 – May 2025**
Mechanical Engineering Intern — *Irvine, CA*

- Supported design reviews and structural analysis of chassis and body systems for high-performance production vehicles.
- Contributed to aerodynamic evaluations and mechanical system analysis. Also applied CAD, physics, and engineering mathematics coursework to real-world design constraints.

Irvine Valley College STEM Club — **Mar 2025 – May 2026**
President — *Irvine, CA*

- Directed club operations and meetings, coordinated industry speaker events, lead hands-on build workshops and technical skill sessions.
- Discussed pathways that developed members' engineering and research fundamentals while connecting them to internship and research pathways.

Irvine Valley College Computer Science Club — **May 2025 – May 2026**
Vice President — *Irvine, CA*

- Managed club operations alongside the President, running weekly speaker sessions, technical workshops, and career development events.
- Lead workshops strengthening member proficiency in different languages, connected students to hackathons, internships, and research opportunities.

Projects

Autonomous Solar-Charged Perimeter Surveillance Drone, Personal Project — **May 2026 - Present**

- Designed and 3D-printed a custom quadcopter airframe from scratch, iterating on frame geometry for structural stiffness and payload capacity.
- Built solar-powered landing stations for autonomous recharge between patrol cycles, integrated into an active residential security system with automated flight routing.

Aircraft Winglet Aerodynamics and Efficiency, IVC / Saddleback Research Symposium — **Aug 2025 - Jan 2026**

- Expanded on my prior award-winning research on winglet performance, rebuilding the study with 3D-printed test articles and an upgraded wind tunnel, then applying machine learning image analysis to characterize wingtip vortex formation across no-winglet, sharklet, and split-scimitar geometries.
- Foundational research earned Second Place in Fluids and Aerodynamics, Senior Division, Orange County Science and Engineering Fair (2021).

Medical Device Research and Development, IVC / Saddleback Research Symposium — **Sep 2024 - Mar 2025**

- Proposed Normothermic Ex-Vivo Heart Perfusion (NEHP) as a testing platform for artificial heart valve validation under physiological flow and pressure conditions.
- Assessed limitations in FDA-approved mechanical validation methods against industry practice, guided by an R&D engineer at Edwards Lifesciences, and demonstrated improved test fidelity using NEHP-perfused porcine hearts.

Biomechanical Shape-Memory Alloy Actuation, OC Science and Engineering Fair — **Nov 2023 - Mar 2024**

- Designed and built a tendon-driven robotic hand using Nitinol shape-memory alloy wire as the actuating element, characterizing actuation response to replicate natural finger articulation.

Additional Information

Skills: SolidWorks, AutoCAD, Fusion 360, Visual Studios, Mechanical Design, Prototyping, 3D Printing, MATLAB, Python, C/C++, JavaScript, Excel, Data Analysis, Git/GitHub, Technical Documentation, SCRUM certified
Interests: Aerospace & Aerodynamics, Autonomous Systems, Robotics & Control Systems, AI in Mechanical Systems, Manufacturing & Production